

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 317: Computer Networks

Batch: 21st
Date: 22/12/2023

Mid-Term Examination

Semester: Fall 2023
Time: 1 Hour 30 Minutes
Total Marks: 30

Answer **any five** questions of the followings.

5 × 6 = 30

1. (a) Define **Computer Networks**. Briefly explain **fault tolerance** and **scalability** of computer networks. 3
(b) What is Distributed Processing? Write the applications of Computer networks. 3
2. (a) Write the full form of **OSI** and **ISO**. 2
(b) Suppose Alice, using a desktop based e-mail client (such as outlook) sends a message to Bob, who accesses his mail using a web-based e-mail account (such as Gmail). Describe how the message gets from Alice's host to Bob's host. 4
- ✓ 3. (a) Why a subnet mask is necessary in IP addressing? Identify the class of the following IP addresses: 2
D i. 11100011.11110000.1100110011.00001111
C ii. C0.AB.12.0 (given in Hexadecimal)
(b) Consider the IP address block, 192.168.10.99/27. **Subnet** the IP block into eight (8) blocks using a fixed length subnet masks (FLSM). Answer the following: 4
i. What is the **fifth** IP address in block 5?
ii. What is the **3rd** usable IP address in block 3?
iii. In which block is the given IP address **192.168.10.99** located?
iv. Write the subnet mask of the FLSM.
- ✓ 4. (a) Consider the subnet mask of a network X.X.X.11000000. Here, X indicates 8 bits of 1. How many valid hosts can be accommodated using this subnet? 2
(b) Suppose you are accessing a local network as Host A. You want to connect to Host B using a layer 2 networking device. Assume for the first time Host A and B connect with each other. Which networking protocol is used for this connection? Describe all the steps for obtaining the MAC address of the destination host. 4
- ✓ 5. (a) What is meant by Loopback Address? What is equal Classless Inter-Domain Routing (CIDR) value of 255.255.248.0 subnet mask? 2
(b) Consider you are given the address space 172.16.10.0/23. Now you are to subnet the address according to the host requirements for different departments that are given in the following table. Then fill up the empty columns with values resulting from your calculation. Use the smallest subnet sizes that will accommodate the relevant number of hosts and do not leave any gap (at the start of given address space or between the subnets). 4

128 64 32 16 8 4 2 1

255.255.254.0

Department	No. of hosts required	Network address	Broadcast address.	Subnet mask
Sales 1	100			...254.0
Service	90			
Accounts	30			
Sales 2	12			
HR	10			

- ✓ (a) People use the Internet for different purposes. Using internet they chat with each other, they doing mathematical problems, view images, watching and listening audio and video files. Which data formats they are accessing in this scenario and how computer represent them? 3
- (b) Write the differences between Circuit Switching and Packet Switching. 3
- ✓ (a) What are the purposes of Dynamic Host Configuration Protocol (DHCP) relay agent? 2
- (b) Explain the messages of DHCP protocol. 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 315 : Microprocessors and Microcontrollers

Batch: 21st

Date: 15/12/2023

Final Examination

Semester: Fall 2023

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions.

5 × 6 = 30

1. (a) What is Microprocessor? Draw the block diagram of a Microprocessor. 3
(b) Write the features of Pentium IV Microprocessor. 3
2. (a) Define the following terms: 3
 - i. Instruction Set
 - ii. Clock Speed
 - iii. Word Length
- (b) Outline the configuration of 8085 Microprocessor. 3
- ✓ (a) Briefly describe the functionality of Program Counter and Accumulator register in 8085 Microprocessor. 3
- (b) Explain the instruction set of 8085 Microprocessor. 3
- ✓ (a) Why 8086 is called a 16 bit Microprocessor? 2
- (b) Briefly Explain the functions of SI, DI, SP, BP and IP. 4
- ✓ (a) What do you mean by Addressing Modes? Mention different Addressing Modes of 8086 Microprocessor. 3
- (b) The value of the Code Segment (CS) Register is 4042H and the value of different offsets is as follows: 3
BX: 2025H , IP: 05F0H , DI: 4247H
Now, calculate the effective address of the memory location pointed by the CS register.
- ✓ (a) What is a Flag Register? Briefly explain the Flag register of 8086 Microprocessor. 3
- (b) Briefly describe SHL and ROL instructions. Also, mention the key differences between ROL and RCL instructions. 3
- ⑦ Draw and describe the Architecture of a 8086 Microprocessor with necessary indications. 6



Process
External Entity
Data store
Data flow

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 313: Database Management System

Batch: 21st

Date: 09/12/2023

Mid-Term Examination

Semester: Fall 2023

Time: 1 hour 30 minutes

Total Marks: 30

Answer any **five** questions.

Marks: 5 x 6 = 30

1. (a) State the responsibilities of a database administrator. 2
(b) Define view. Compare among different types of views. 4
2. (a) Write the guidelines for creating an Index. 2
(b) Briefly explain the rules performing and denying the DML operations on a view. 4
3. (a) Define Context diagram with an example. 2
(b) State the guidelines for drawing a DFD. 4
4. (a) State the roles and skills of a system analyst. 3
(b) Briefly explain level-0 and level-1 DFD with example. 3
5. Find the Data Flow Diagramming (DFD) rules. 6
6. Design an E-R Diagram for the following scenario: 6

Employee
Employee_ID
Employee_Name
Employee_Gender
Employee_Address
Employee_DoB
Employee_DoJ

Department
Dept_ID
Dept_Name
Dept_Location
Dept_Phone

Project
Proj_ID
Proj_Name
Proj_Location
Proj_Duration

Dependent
Depn_Name
Depn_Gender
Depn_relation

7. By using the scenario of Question 6, Implement the SQL command to create all the relations. Write the SQL to create a view name **empvu** containing all the columns from the Employee table. 6



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE311: Computer Architecture

Batch: 21st

Date: 08/12/2023

Mid-Term Examination

0.025000

Semester: Fall 2023

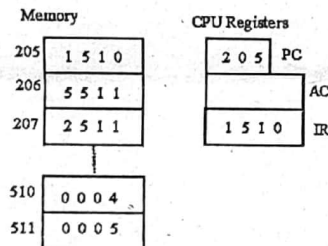
Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions of the followings.

5 × 6 = 30

- ✓ (a) Briefly explain the techniques used to speedup microprocessor. 3
- (b) Draw the top-level view of computer system. 3
2. (a) Suppose, a processor has access to three levels of memory. Level 1 contains 1500 words and has an access time of 0.02 μ s; level 2 contains 50000 words and has an access time of 0.1 μ s; level 3 contains 1000000 words and has an access time of 0.25 μ s. If 70% of the memory accesses are found in level 1 and 24% of the memory accesses are found in level 2, then calculate the average time to access a word. 2
- (b) Draw and Explain Instruction cycle state diagram. 4
- ✓ 3. (a) Illustrate priority interrupt with an example. 3
- (b) Complete the following program execution where contents of memory and registers given in hexadecimal. 3



4. (a) Describe Set-associative mapping for cache memory with an example. 3
- (b) Draw the block diagram of Pentium-4. 3
- ✓ Consider the execution of a program that results in the execution of 3000000 instructions on a 404-MHz processor. The program consists of four major types of instructions. The instructions mix and the CPI for each instruction type are given below based on the result of a program trace experiment: 6

Instruction Type	CPI	Instruction Mix (%)
Arithmetic and logic	1	55
Load/store with cache hit	2	20
Branch	4	18
Memory reference with cache miss	8	7

The result of CPI = 2.23 and MIPS rate = 181. Now assume that the program can be executed in eight parallel tasks or threads with roughly equal number of instructions executed in each task. Execution is on an 8-core system with each core (processor) having the same performance as the single processor originally used. Coordination and synchronization between the parts adds an extra 25,000 instruction executions to each task. Assume the same instruction mix as in the example for each task, but increase the CPI for memory reference with cache miss to 12 cycles due to contention for memory.

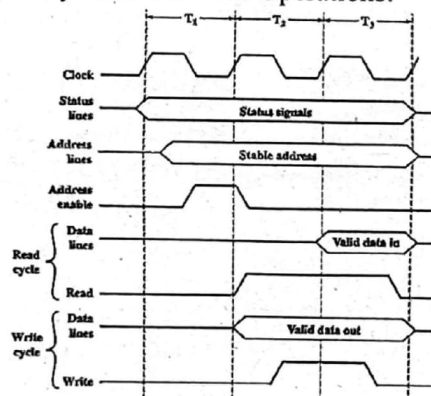
- Determine the average CPI and corresponding MIPS rate.
- Calculate the speedup factor.
- Compare the actual speedup factor with the theoretical speedup factor determined by Amdahl's law.

6. ~~(a)~~ Explain the write through and write back techniques used in cache write process. 2

(b) Consider a machine with a byte addressable main memory of 216 bytes and the block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines is used with this machine. 4

- How is a 16-bit memory address divided into tag, line number, and byte number?
- Into what line would bytes with each of the following addresses be stored?
0001 0001 0001 1011
1100 0011 0111 0100
1101 0000 0101 1101
1010 1010 1010 1010
- Suppose the byte with address 0001 1010 0001 1010 is stored in the cache. What are the addresses of the other bytes stored along with it?
- How many total bytes of memory can be stored in the cache?

7. Consider the following timing of Synchronous Bus Operations:



- Assume the microprocessor bus is clocked at 10 MHz and that the Read signal begins to fall in the middle of the second half of T₃: 3
 - Determine the length of the memory read instruction cycle.
 - When should memory data be placed on the bus? Allow 20 ns for the settling of data lines.
- Assume the microprocessor read bus timing requires four processor clock cycles. The valid data is on the bus for an amount of time that extends into the fourth processor clock cycle. Assume a processor clock rate of 8 MHz. 3
 - What is the maximum data transfer rate?
 - Repeat but assume the need to insert one wait state per byte transferred.



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering MAT317: Engineering Mathematics

Semester: Fall 2023

Time: 1 Hour 30 Minutes

Total Marks: 30

23-12-2023

Midterm Examination

Answer any 3(Three) out of the following questions

10×3=30

- ✓ 1. a. Discuss about the power series method 3
b. Show that $x = 0$ is a regular singular point of the differential equation 7
- $$3x \frac{d^2 y}{dx^2} - (x - 2) \frac{dy}{dx} = 0$$
- also find a Frobenius type series solution of this differential equation
2. a. Define Fourier series. Write the Dirichlet's conditions of Fourier series. 4
b. Expand the function $f(x) = x + x^2$ in Fourier series in the interval $-\pi \leq x \leq \pi$ 6
- and hence deduce that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$
3. a. Discuss about Fourier sine and cosine transformation. 5
b. Define Fourier integral. Using Fourier integral evaluate $\int_0^{\infty} \frac{\sin ux}{u^2 + 1} du, x > 0$ 5
- ✓ 4. a. Define Laplace transform with examples. Write the properties of Laplace transform. 3
b. Show that $L\{t^n\} = \frac{n!}{s^{n+1}}$ 5
- ✓ 5. a. Evaluate $L\{t(\sin at + e^{at})\}$ 2
b. Write the convolution property of inverse Laplace transform. By this property 3
evaluate $L^{-1}\left\{\frac{1}{(s+1)^3 s^2}\right\}$
- b. With the help of Laplace transform solve: 3
- $$\frac{d^2 y}{dt^2} + 9y = \cos 2t, \quad y(0) = 0, y\left(\frac{\pi}{2}\right) = -1.$$
- c. With the help of Laplace's transform solve: 4
- $$\frac{dx}{dt} - y = e^t$$
- $$\frac{dy}{dt} + x = \sin t, \text{ Where } x(0) = 0, y(0) = 1$$

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 317: Computer Networks

Semester: Fall 2023

Time: 2 Hours

Total Marks: 40

Batch: 21st
 Date: 18/03/2024

Final Examination

Answer any **five** questions of the followings.

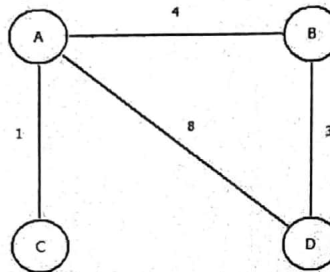
5 × 8 = 40

1. (a) What is the relation between Domain Name Server (DNS) and IP addressing? 4
 Briefly explain the hierarchy of DNS.

(b) A host 'du.ac.edu' wants to find the IP address of 'hamdarduniversity.edu.bd/cse/faculty'. 4
 Illustrate how the DNS resolution works with an explanation.

2. (a) Define distance vector routing algorithm. 2

(b) Consider the following diagram: 6



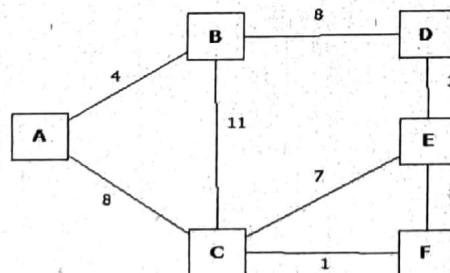
Calculate the shortest path using the Bellman-ford algorithm for this diagram.

3. (a) Explain the Internet Control Message Protocol (ICMP) message formats. 3

(b) Write the names of error reporting messages. Briefly explain the type 4 and type 11 error reporting messages. 5

4. (a) Write the differences between distance vector and link state routing algorithm. 3

(b) Find the shortest path for the following network of router C using the link state routing protocol: 5

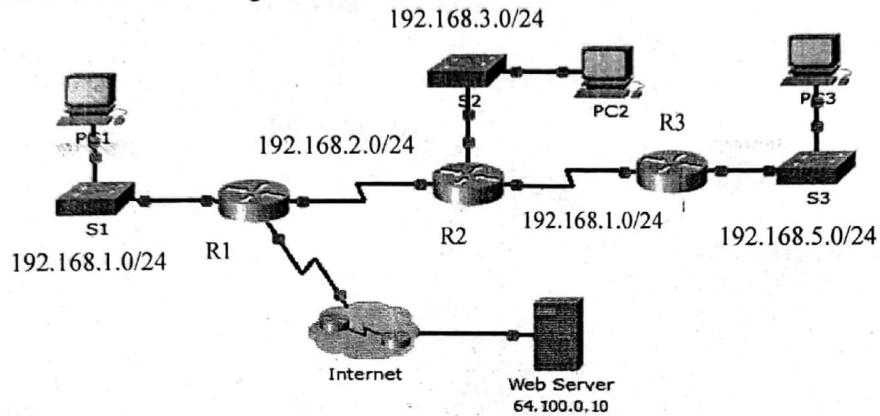


CFEADB

5. (a) Define single-area and multi-area Open Shortest Path First (OSPF). Briefly explain the procedures to maintain the routing information of OSPF. 4

(b) Explain the message formats of OSPF. 4

- ✓ 6. (a) What is a port address? Briefly explain the client-server programming. 3
(b) Define Network Address Translation (NAT). How many types of addresses in NAT? Briefly explain all addresses. 5
7. (a) What is Routing Information protocol in computer networks? 2
(b) Consider the following network: 6



Configure the RIPv2 for the router R1, R2 and R3 with commands.

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 315 : Microprocessors and Microcontrollers

Batch: 21st

Semester: Fall 2023

Date: 13/03/2024

Final Examination

Time: 2 Hour

Total Marks: 40

Answer any **five** questions.

5 × 8 = 40

1. (a) What is Microcontroller? Mention the structural components of a Microcontroller. 2
2. (b) Describe the different types of Microcontroller with necessary figures. 6
3. (a) Differentiate between CISC and RISC Processors. 3
4. (b) Write the High Performance CPU and Peripheral features of PIC16F877A. 5
5. (a) Briefly describe —
I. USART
II. SPI
III. I2C 3
6. (b) Draw the architecture of a PIC Microcontroller and describe its memory organization. 5
7. (a) Write the differences between Microcontroller and Microprocessor. 4
8. (b) State the steps for EEPROM Write in PIC16F877A. 4
9. (a) Define RTOS. Describe the components of an RTOS. 4
10. (b) What are interrupts in computer architecture? Briefly explain the interrupt cycle with necessary figures. 4
11. (a) How can the processor recognize which device has requested the interrupt? 3
12. (b) Explain the working principle of Daisy Chaining method with necessary figures. 5
13. (a) What is Interrupt Latency? 2
14. (b) Distinguish between closely coupled and loosely coupled microprocessor configurations with necessary figures. 6

M.C → RAM → ROM
SI → Timmer → I/O ports
SFR RAM
Program Memory
EEPROM
execute



Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 313: Database Management System

Batch: 21st

Date: 10/03/2024

Final Examination

Semester: Fall 2023

Time: 2 hours

Total Marks: 40

5 x 8 = 40

Answer any **five** questions.

✓ Let, we have the following **Customer** table

Customer

Cus_id	Cus_name	Cus_email	Cus_contact	Cus_addr	Cus_bg	Cus_dept
120223001	Afia Khan	a@gmail.com	01913000000	Dhaka	A +ve	CSE
120223002	Rabeya	b@yahoo.com	01721111111	Sonargaon	O +ve	CSE
120223003	Siam	c@hub.edu.bd	01821222222	Dinajpur	B +ve	EEE
120223004	Sadia	d@gmail.com	01911333333	Narayanganj	O +ve	Econ
120223005	Siam	e@hub.edu.bd	01551444444	Munshigonj	O -ve	BBA

- (a) Using SQL, create a table Cus_CSE(Cus_id, Cus_name, Cus_email, Cus_contact, Cus_addr, Cus_bg) from the above Customer table. 2
- (b) Derive the SQL to insert Siam's Cus_dob = '05-05-1995' in the EEE department of the above Customer table. 2
- (c) Write the SQL to insert the following data into the above Customer table:
Data: 120223007, 'Abu Toub', 'aabbcc@hub.edu.bd', 01813123456, 'Dhaka', 'A+ve', '11-01-1995'. 4
- ✗ (a) Write the properties of relational databases. 2
- (b) State the differences between SQL and SQL *Plus. 2
- (c) Write the SQL for creating a sequence, modifying a sequence. 4
3. (a) How index created? State the guidelines for creating index. 3
- (b) Draw an E-R diagram for the Registration system of Hamdard University Bangladesh. 5
- ✗ (a) Briefly explain the function of a 3-Tier Client-Server Database Architecture with diagram. 4
- (b) Briefly explain the concurrency control and recovery with merits and demerits of a distributed database system. 4
5. (a) Define subquery. Write the guidelines for using subqueries. 3
- (b) Let we have the table **employees**(employee_id, first_name, last_name, email, phone_no, hire_date, job_ID, salary, manager_ID, Dept_ID). Now write the SQL to determine the followings: 5
 - i) Show the first_name and last_name of the employee who has a greater salary than Ahmed's salary by using subquery.
 - ii) Show the last_name, job_ID and salary of the employee whose Job_ID not only containing employee ID 141 but also having the greater salary containing the employee_ID 143 by using subquery.

Find the Cartesian product operation for the following relations:

Std

Std_id	Std_name	Std_contact
315151001	Albance	123456789
315151002	Balbao	123456788
315151003	Cheraio	123456787
315151004	Dernaio	123456786

Contact

Std_contact	Std_addr	Std_bg
123456789	Dhaka	A +ve
123456788	Sonarga	AB +ve
123456787	Voberch	B+ve
123456786	Gazipur	O +ve

(b) Determine different types of Outer-Join and write the SQL for the following relations:

Std

Name	Age	Course
Afia	22	DBMS
Sadia	24	CP&I
Siam	23	BAN
Toub	21	ENS

Subject

Course	Day
AI	Monday
CP&I	Tuesday
DBMS	Wednesday
MicroP	Thursday
Network	Friday

Derive the Comparison on 3NF and BCNF.

Consider the following **Student** table with variable repeating groups. Find 1NF, 2NF and 3NF for the given Student table.

Student

Std_Id	Adm_Date	Std_name	Std_email	Grade	Cr_Points	Course_Name	Ob_Mark
31501	11/03/18	Dev Joe	dev@iewr.com	A	1	Discrete	76
31501	11/03/18	Dev Joe	dev@iewr.com	A	1	Algorithms	77
31501	11/03/18	Dev Joe	dev@iewr.com	A	1	Fundamentals	76
31502	12/03/18	Dany Sam	dany@ter.com	B	4	Algorithms	66
31502	12/03/18	Dany Sam	dany@ter.com	B	4	Fundamentals	67



Answer any **four** questions of the followings.

5 × 8 = 40

- ✓ (a) Draw the architecture of a typical 16 Megabit DRAM. 4
- ✓ (b) The memory of a particular microcomputer is built from 64K × 1 DRAMs. According to the data sheet, the cell array of the DRAM is organized into 256 rows. Each row must be refreshed at least once every 8ms. Suppose we refresh the memory on a strictly periodic basis. 4
- M
- i. What is the time period between successive refresh requests?
- ii. How long a refresh address counter do we need?
- ✓ (a) Draw the Block Diagram of an I/O Module. 2
- ✓ (b) In virtually all systems that include DMA modules, DMA access to main memory is given higher priority than CPU access to main memory. Why? 2
- ✓ (c) Illustrate the Instruction Cycle State Diagram. 0 4
- ✓ (a) Briefly explain all the level of Redundant Array of Independent Disks (RAID). 4
- ✓ (b) Consider a disk that rotates at 3600 rpm. The seek time to move the head between adjacent tracks is 2ms. There are 32 sectors per track, which are stored in linear order from sector 0 through sector 31. The head sees the sectors in ascending order. Assume the read/write head is positioned at the start of sector 1 on track 8. There is a main memory buffer large enough to hold an entire track. Data is transferred between disk locations by reading from the source track into the main memory buffer and then writing the data from the buffer to the target track. 4
- M
- i. How long will it take to transfer sector 1 on track 8 to sector 1 on track 9? 64 ms
- ii. How long will it take to transfer all the sectors of track 8 to the corresponding sectors of track 9?
- ✓ (a) Illustrate Interrupt-Driven I/O operations. 4
- ✓ (b) Draw and explain how System Bus use in 8237 DMA. 4
5. (a) If an instruction contains four addresses, what might be the purpose of each address? 2
- (b) Define read hit and read miss. 2
- (c) Explain Indirect addressing and Register addressing with proper diagram. 4
6. (a) Briefly explain the Common fields of program status word. 2
- (b) Draw the Six-Stage CPU Instruction Pipeline. 2
- (c) What is Loop Buffer? Write the benefits of it. 4

- ✓ 7. (a) Explain the design issues in multiprocessor Operating System. 4
- (b) **Clustering** is an alternative to symmetric multiprocessing as an approach to providing high performance and high availability and is particularly attractive for server applications. What are the design requirements of **clustering**? 4



HAMDARD UNIVERSITY BANGLADESH

Final sem

Department of Computer Science and Engineering

MAT317: Engineering Mathematics

Semester: Fall 2023

Time: 2 Hours

23-03-2024

Final Examination

Total Marks: 40

Answer any 4(Four) out of the following questions

10×4=40

1. a. State and prove generating function for Bessel's function $J_n(x)$. 5
 b. Show that $\left[J_{\frac{1}{2}}(x)\right]^2 + \left[J_{-\frac{1}{2}}(x)\right]^2 = \frac{2}{\pi x}$. 5
2. Derive the Legendre polynomials $P_n(x)$ for Legendre differential equation. 10
3. a. State and prove the Rodrigue's formula for Hermite polynomials. 5
 b. Find the relations between Hermite and Legendre polynomials. 5
4. Solve the boundary value problem $\frac{\partial^2 y}{\partial t^2} = 16 \frac{\partial^2 y}{\partial x^2}$, $0 < x < 2, t > 0$ subjected to the conditions $y(0, t) = 0, y(2, t) = 0, y(x, 0) = 6 \sin \pi x - 3 \sin 4 \pi x$ and $y_t(x, 0) = 0$. 10
5. a. Discuss about Sturm-Liouville problem, Eigen function and Eigen value of Sturm-Liouville problem with examples. 3
 b. Find the Eigen function and Eigen value of the boundary value problem 7

$$y'' + \lambda y = 0; y(-1) = y(1), \quad y'(-1) = y'(1).$$
6. Construct the Green's function and solve $y'' + y = 1, y(0) = 0, y(1) = 0$. 10

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r \left(\frac{x}{2}\right)^{n+2r}}{\sqrt{r+1} \sqrt{n+r+1}}$$